A CASE-STUDY BASED REPORT

ON

CUSTOMER PURCHASE PREDICTION USING LOGISTIC REGRESSION AND RANDOM FOREST

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MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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Abstract

Customer Purchase Prediction Using Logistic Regression and Random Forest is a machine-learning case study that estimates whether a customer is likely to purchase using demographic and behavioural information. The dataset contains 5,000 customer records and 13 columns, including age, gender, annual income, customer segment, previous purchases, website visits, pages viewed, time spent on the website, emails opened, cart additions and days since the last purchase. The original dataset was checked for missing values and duplicate records; 30 duplicate rows and 45 missing cells were identified. After duplicate removal, 4,970 records remained. Customer_ID was excluded from modelling because it is an identifier. Numerical and categorical missing values were treated using median and most-frequent imputation, respectively, while encoding and scaling were implemented in a Scikit-Learn preprocessing pipeline. Logistic Regression and Random Forest were evaluated, and Random Forest hyperparameters were explored using 5-fold GridSearchCV. K-Means, the Elbow Method, Silhouette Score, DBSCAN and Hierarchical Clustering were also used to explore customer groups. The reported test accuracies were 61.77% for Logistic Regression, 57.95% for Random Forest and 59.86% for Tuned Random Forest.

1. Introduction

1.1 Background

Customer purchase prediction is a supervised machine-learning task that estimates whether a customer will purchase a product or service. Retail and e-commerce businesses collect demographic, purchase-history and website-engagement information. Analysing these attributes can help identify patterns associated with purchasing behaviour. This case study uses Logistic Regression and Random Forest to classify customer records into purchase and no-purchase outcomes.

1.2 Motivation for the Project

The project demonstrates an end-to-end machine-learning workflow, including data-quality checks, preprocessing, exploratory data analysis, customer clustering, classification, hyperparameter tuning and evaluation. It also compares a linear baseline with a tree-based ensemble and highlights the importance of evaluating precision, recall, F1-score and ROC-AUC rather than relying only on accuracy.

1.3 Overview of the Work Carried Out

The CSV dataset was loaded and inspected. Duplicate records and missing cells were identified, and Customer_ID was excluded from the model inputs. Numerical and categorical data were imputed, categorical features were one-hot encoded, and numerical features were standardized. EDA was performed using descriptive statistics, distribution plots, boxplots and a correlation heatmap. K-Means, DBSCAN and hierarchical clustering were used for unsupervised exploration. Logistic Regression and Random Forest were trained using a stratified 80:20 train-test split. GridSearchCV with five-fold cross-validation was used to tune Random Forest. Models were evaluated using accuracy, precision, recall, F1-score, confusion matrices and ROC-AUC.

1.4 Scope and Responsible-Use Statement

This study is limited to the supplied dataset and its available features. Model outputs represent statistical estimates, not certainty about an individual customer, and the observed associations do not establish causation. Before real-world use, performance on newer data, privacy, fairness, probability calibration and business consequences should be evaluated. Predictions should support responsible decision-making rather than replace human judgement.

2. Problem Statement

2.1 Case-Study Scenario

An online retail business wants to estimate whether a customer is likely to purchase based on demographic and website-behaviour attributes. Historical records contain customer features and a binary Purchase label. A classifier is required to learn from the records and predict the label for a new customer.

2.2 Central Case-Study Question

Can demographic and behavioural customer features predict the purchase outcome, and how do Logistic Regression and Random Forest compare on a held-out test set?

2.3 Formal Problem Definition

Let X represent a customer's input features and y represent Purchase, where 0 means No Purchase and 1 means Purchase. The task is to learn a classifier f(X) that predicts the label for previously unseen records. The same preprocessing transformations learned during training must be applied at prediction time.

2.4 Objectives that Guide the Study

Describe the customer dataset, input features and target variable.

Identify and remove duplicate records and handle missing values.

Exclude Customer_ID from modelling.

Perform EDA using descriptive statistics, plots, boxplots and a correlation heatmap.

Explore customer groups using K-Means, Elbow Method, Silhouette Score, DBSCAN and Hierarchical Clustering.

Train Logistic Regression and Random Forest with a leakage-aware preprocessing pipeline.

Tune Random Forest using five-fold GridSearchCV.

Compare models using multiple classification metrics.

Demonstrate prediction for a new customer record.

2.5 Assumptions and Limitations

Results depend on the dataset, split, preprocessing and model configuration. Purchase-class recall is relatively low in the reported runs, meaning that some actual purchasers were classified as non-purchasers. Scores may not generalize to other customers or future periods. Cluster assignments are exploratory and should not be interpreted as verified customer types.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The CSV dataset contains 5,000 rows and 13 columns. Customer_ID is an identifier, Purchase is the target, and the remaining 11 columns are candidate predictors.

Property	Value
Initial records	5,000
Initial columns	13
Target	Purchase (0 = No Purchase; 1 = Purchase)
Identifier excluded	Customer_ID
Duplicate rows removed	30
Records after duplicate removal	4,970
Missing cells identified in original dataset	45
Supervised predictors	11

The recorded original class counts are 2,921 No Purchase and 2,079 Purchase. After duplicate removal, the reported cleaned class counts are 2,905 No Purchase and 2,065 Purchase, totalling 4,970 records.

The 11 predictors are Age, Gender, Annual_Income, Customer_Segment, Previous_Purchases, Website_Visits, Pages_Viewed, Time_on_Website_Min, Emails_Opened, Cart_Additions and Days_Since_Last_Purchase.

3.2 Purpose of EDA

EDA was used to understand class distribution, feature distributions and relationships before modelling. Descriptive statistics and plots were used to inspect numerical variables; categorical plots examined Gender and Customer_Segment; boxplots compared numerical values across purchase outcomes; and a correlation heatmap examined linear relationships between numerical features and Purchase.

3.3 Preparation Steps Actually Performed

Loaded the CSV into Pandas and inspected its dimensions and columns.

Checked duplicates and removed 30 rows, leaving 4,970 records.

Identified 45 missing cells in the original dataset.

Removed Customer_ID from supervised model inputs.

Used median imputation for numerical features and most-frequent imputation for categorical features.

Applied OneHotEncoder to categorical features and StandardScaler to numerical features.

Used a stratified 80:20 train-test split with random_state=42.

Kept preprocessing in a Scikit-Learn Pipeline/ColumnTransformer to learn transformations from training data.

Pandas normally interprets blank CSV cells and literal NaN entries as missing values when the CSV is loaded. They are counted together for missing-value treatment.

3.4 EDA Exploration - Implementation

The EDA implementation included descriptive statistics, a purchase-class distribution chart, numerical and categorical feature distributions, boxplots and a correlation heatmap. The original class distribution was 2,921 No Purchase and 2,079 Purchase records. After duplicate removal, 2,905 No Purchase and 2,065 Purchase records remained.

The reported correlation analysis showed generally weak linear correlations between individual numerical features and Purchase. This suggests that no single numerical feature had a strong linear relationship with the target in the inspected data; it does not rule out combined or nonlinear relationships.

3.5 Summary of EDA Findings

The dataset contains demographic and website-behaviour features.

The target variable is binary.

Duplicate removal reduced the dataset from 5,000 to 4,970 records.

Missing values required numerical and categorical imputation.

Individual numerical features showed generally weak linear correlations with Purchase.

EDA informed preprocessing and model development but did not establish causal effects.

4. Algorithm Selection and Implementation

4.1 Preprocessing Pipeline

A ColumnTransformer applied different transformations to numerical and categorical columns. Numerical features were median-imputed and standardized with StandardScaler. Categorical features were imputed using the most-frequent value and encoded with OneHotEncoder(handle_unknown='ignore'). These transformations were combined with each classifier in a Scikit-Learn Pipeline. Fitting the pipeline on training data helps prevent preprocessing leakage into the test set.

4.2 Logistic Regression

Logistic Regression was selected as a linear baseline for binary classification. It estimates the probability of the positive class and uses the configured decision threshold to produce a class label. The reported pipeline used max_iter=1000 and random_state=42.

4.3 Random Forest

Random Forest combines multiple decision trees and can model nonlinear patterns and interactions between features. The reported baseline used n_estimators=100 and random_state=42. Feature importance was examined to describe the fitted model; importance scores should not be interpreted as causal effects.

4.4 Unsupervised Learning

Unsupervised learning explored customer groupings without using Purchase as an input to clustering. Numerical features were imputed and standardized before distance-based clustering. These results are exploratory and separate from supervised purchase classification.

4.4.1 K-Means and Elbow Method

K-Means was applied to standardized numerical customer features. The Elbow Method compared inertia for K values from 2 to 8. The inertia curve was examined for a point where adding clusters produced diminishing reductions in within-cluster variation.

4.4.2 Silhouette Score

Silhouette Score was calculated for tested K values to assess cluster cohesion and separation. The reported highest tested score was approximately 0.0837 at K=2. This low score suggests limited separation between the tested clusters, so the grouping should be treated as exploratory.

4.4.3 DBSCAN

DBSCAN was run with eps=1.5 and min_samples=5 on standardized clustering features. The reported output identified 32 clusters and 1,353 observations labelled as noise/outliers. These results depend on scaling and parameter choices.

4.4.4 Hierarchical Clustering and Dendrogram

Agglomerative hierarchical clustering was performed using Ward's linkage, and a dendrogram was generated to visualize the order in which groups merged at different distances. A final number of clusters should only be reported if a dendrogram cut level was explicitly selected.

4.5 Hyperparameter Tuning

GridSearchCV was applied to the Random Forest pipeline using five-fold cross-validation and accuracy as the scoring metric. The search considered n_estimators, max_depth and min_samples_split. The reported best cross-validation accuracy was 60.89%, with n_estimators=50, max_depth=10 and min_samples_split=5. Verify these values against the latest notebook output before submission.

4.6 Evaluation Metrics

Accuracy is the proportion of correctly classified records. Precision measures the reliability of positive predictions, while recall measures the proportion of actual purchasers identified. F1-score combines precision and recall. Confusion matrices show true negatives, false positives, false negatives and true positives. ROC-AUC measures the model's ability to rank positive cases above negative cases across thresholds.

4.7 Classification Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	61.77%	57.02%	32.45%	41.36%	0.6351
Random Forest	57.95%	48.90%	26.88%	34.69%	0.5775
Tuned Random Forest	59.86%	53.40%	26.63%	35.54%	0.6046

Logistic Regression achieved the highest reported test accuracy and F1-score among the three models. Tuned Random Forest reached 60.89% cross-validation accuracy and 59.86% test accuracy. All three models had relatively low recall for the Purchase class, indicating that some actual purchasers were missed.

4.8 Confusion Matrix Results

Model	Confusion matrix	Interpretation
Logistic Regression	[[480, 101], [279, 134]]	480 true negatives, 101 false positives, 279 false negatives, 134 true positives.
Random Forest	[[465, 116], [302, 111]]	465 true negatives, 116 false positives, 302 false negatives, 111 true positives.

Both matrices show more correctly classified No Purchase records than Purchase records. The false-negative counts indicate that many actual purchasers were predicted as non-purchasers. Verify these matrices against the final notebook if the models have been rerun.

4.9 Final Prediction Demonstration

The demonstration record used Age=35, Gender=Male, Annual_Income=80000, Customer_Segment=Regular, Previous_Purchases=5, Website_Visits=10, Pages_Viewed=6, Time_on_Website_Min=12, Emails_Opened=4, Cart_Additions=2 and Days_Since_Last_Purchase=60.

The reported Logistic Regression prediction was Purchase=0 (No Purchase), with a predicted purchase probability of 0.3504 (35.04%). This is a model estimate for the given record, not a guarantee of the customer's behaviour. Retain this result only if it matches the latest notebook output.

5. Conclusion

This case study applied a machine-learning workflow to customer purchase prediction using a dataset containing 5,000 records and 13 columns. The process included duplicate removal, missing-value treatment, identifier exclusion, encoding, scaling and EDA. K-Means, the Elbow Method, Silhouette Score, DBSCAN and Hierarchical Clustering were used to explore customer groupings. Logistic Regression and Random Forest were trained for supervised classification, and Random Forest hyperparameters were explored using five-fold GridSearchCV. Logistic Regression achieved 61.77% test accuracy and a 41.36% F1-score; Random Forest achieved 57.95% accuracy; and Tuned Random Forest achieved 59.86% accuracy. The relatively low recall for purchasers indicates that further work could explore class weighting, calibration, decision thresholds, additional features and validation on newer data. The project demonstrates data preparation, clustering, classification, evaluation and prediction while recognizing the limitations of the dataset and models.